



Linear mixed effects models

Journal Club
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Overview

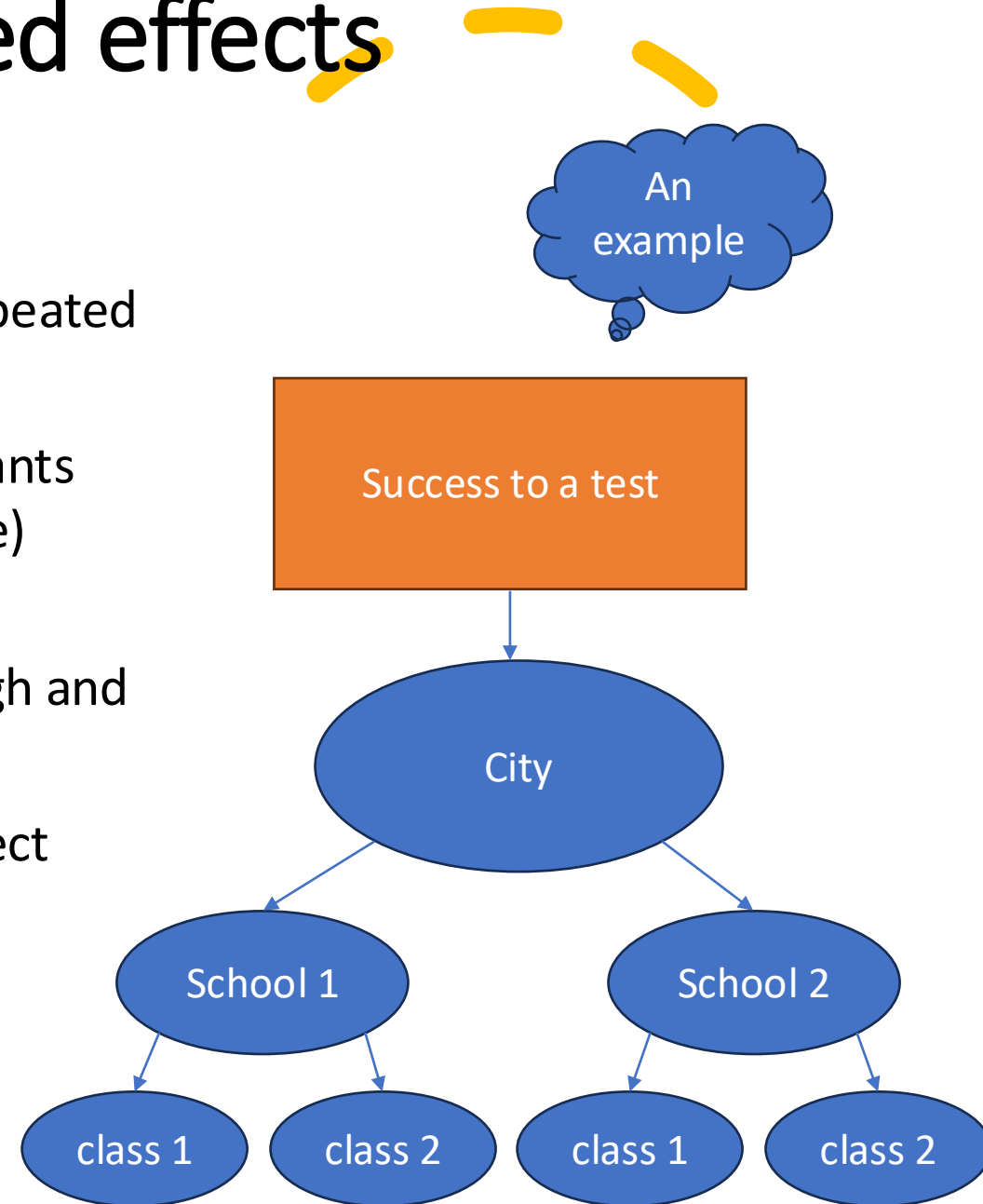
- Background
 - ANOVAs vs Mixed models
 - Fixed effects and random effects
 - How do I do it?
 - Maths
- Real world examples
- Common things you might want to do
 - Power analysis
 - Model selection
 - More complicated mixed models



Background

Why should we use linear mixed effects models rather than ANOVAs?

- ✓ LME are basically just ANOVAs – they can both look at repeated measures data
- ✓ LMEs can account for variability within/between participants and items at the same time (ANOVAs can only look at one)
- ✓ LMEs deal with missing data well (if one case is missing ANOVAs delete the entire case) – if dataset is large enough and needs to be random
- ✓ LMEs can tell us about the magnitude/direction of an effect (ANOVAs can't)
- ✓ LMEs can deal with categorical or continuous variables (although the analysis differs)
- ✓ More flexible for hierarchical data



Why are they called mixed effects models?

- LMEs are mixed models – this means that we model both fixed and random effects
 - Fixed = population-level (average) effects that should persist across experiments. If the effect will persist in different sample then it is considered fixed.
 - These are the effects you want to study
 - Can be continuous or categorical
 - E.g. task
 - Random = effects that may differ from average trend (i.e. across ppt or items)
 - These are variations you want to discard
 - They are categorical (because they are discrete units samples from the same population) so if you don't know whether an effect should be random/fixed – it cannot be random if it is continuous.
 - E.g. subject, or item
 - (you can transform your continuous random effect into a categorical one if necessary)

I want more maths...

- An LME is expressed as a math equation:

$$y = X\beta + Zu + \varepsilon$$

With y the outcome variable, X the predictors (e.g. test, group) and β the associated fixed effects, Z the matrix of u random effects and ε the error.

- The result is on the left side of the equal sign. Then comes the fixed effects, the random effects and finally the error factor.
- Complex models include multiple fixed and random effects, nested random effects, fixed effects with interaction...

- For this example: $y = X\beta + Zu + \varepsilon$
- An outcome y for the visual acuity (decimal Snellen): $0.01 < y < 2.0$

- Three X predictors:

<i>Intercept</i>	<i>Stargardt</i>	<i>Age</i>
1	0	30
...	...	...
1	1	25

- With their β fixed effects:

<i>Intercept</i>	<i>Stargardt</i>	<i>Age</i>
0.15	-0.4	-0.01

- And the random effect matrix (not shown)

I'm interested... how do I do it?

- A common package in R is called “lme4”, although there are multiple that you can use
 - Lme4 is easy to use, fast but does not supply p values.
 - Other packages can be used in combination to get the p value e.g. lmerTest/lmerstat
- Basic model in R
 - `lmer(outcome ~ 1 + predictor + (1|participant) + (1|item), data=data)`
 - `( )` = random effects
 - Vertical lines are pipes – the effects on the left vary by the grouping factor of the right – left cannot be left blank – they recommend always using 1 in both fixed and random effects to avoid confusion about when it should be used
 - Can also add random slope by `(1+predictor|item)`

Multiple types of random effects

Formula	Alternative	Meaning
$(1 \mid g)$	$1 + (1 \mid g)$	Random intercept with fixed mean.
$0 + \text{offset}(o) + (1 \mid g)$	$-1 + \text{offset}(o) + (1 \mid g)$	Random intercept with <i>a priori</i> means.
$(1 \mid g1/g2)$	$(1 \mid g1) + (1 \mid g1:g2)$	Intercept varying among $g1$ and $g2$ within $g1$.
$(1 \mid g1) + (1 \mid g2)$	$1 + (1 \mid g1) + (1 \mid g2).$	Intercept varying among $g1$ and $g2$.
$x + (x \mid g)$	$1 + x + (1 + x \mid g)$	Correlated random intercept and slope.
$x + (x \parallel g)$	$1 + x + (1 \mid g) + (0 + x \mid g)$	Uncorrelated random intercept and slope.

Model selection

- Remember, a simpler model is (almost) always better to find out which model is the best for your data. BUT sometimes we want to see if one model returns significant results
- Likelihood ratios
 - Create two models – one reduced (basic) and one complex
 - Use `anova()` to determine which has a better fit to the data
 - `anova(rt_reduced.mod, rt_full.mod)`
 - Can also use the package “afex” which will do this for you – means that you don’t accidentally use the reduced models
 - `mixed (outcome ~ 1 + predictor + (1|participant) + (1|item), data=data)`
- Stepwise regression
 - Controversial – some people really hate these



Power Analysis / Sensitivity analysis

Two sides of the same coin. Journals now require one or the other.

If you have previous or pilot data, then you can initiate a power analysis, using the effect sizes you noticed in your pilot data and estimate the number of people you need for 80% power.



If you have no previous or pilot data and the dataset is already done, you can initiate a sensitivity analysis using the actual parameters of your data (N, design, random/fixed effects) and determine the minimal effect size required for 80% power. If your actual effect size is higher, your mixed model is justified.



Power analyses should ONLY be done before your analysis. Sensitivity analyses can be done at any step of the analysis.



Common things
you might want to
do

More about mixed models

- Models with categorical fixed effects ONLY
 - The results from a model only give partial information. You need to run several models for all factors of the categorical fixed effects to get a definitive answer. However, doing only one factor does not change the post-Hoc analysis
- Bayesian mixed models
 - The bmrs package on R is very similar to lme4 (similar syntax in model formula and results)
 - Same principle
 - Harder way to make your power/sensitivity analysis (see https://bookdown.org/ajkurz/DBDA_recoded/goals-power-and-sample-size.html)
 - In terms of interpretation, way more useful. Direction and Magnitude of the effects can be discussed whether the 'significance' of it.

How do I report my results?

- No explicit rules
- They usually report
 - χ^2 – likelihood ratio
 - Degrees of freedom of the test
 - Associated p-value
 - Coefficient estimates, t values and standard errors associated with parameters of interest
 - For example:
 - A likelihood-ratio test indicated that the model including modality provided a better fit for the data than a model without it, $\chi^2(1) = 32.39$, $p < .001$. Examination of the summary output for the full model indicated that response times were on average an estimated 83 ms slower in the audiovisual relative to the audio-only condition ($\beta = 83.18$, $SE = 12.58$, $t = 6.62$).
- Lme4 doesn't report p values so will need to use another package (e.g. lmerTest)
- Or...Hanna's new favourite r package:
 - `report(vpl_blind_fa_model)`

How do we compare across studies?

- We use standardized beta coefficients instead of effect sizes
 - <https://www.statisticshowto.com/standardized-beta-coefficient/>:
 - A standardized beta coefficient compares the strength of the effect of each individual independent variable to the dependent variable. The higher the absolute value of the beta coefficient, the stronger the effect.
 - For example, a beta of $-.9$ has a stronger effect than a beta of $+.8$.
 - Standardized beta coefficients have standard deviations as their units.
 - This means the variables can be easily compared to each other.
 - In other words, standardized beta coefficients are the coefficients that you would get if the variables in the regression were all converted to z-scores before running the analysis

How to start?

We made an R markdown that describe everything and uses data examples! (simulated data and data that was gathered at our lab)

COMING SOON



References / More info

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